

Veterans Affairs Network Security Operations Center

Visibility Team



NetScaler Application Pool Monitoring Configuration Standard Operating Procedure

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1. Introduction

The Department of Veterans Affairs Network Security Operations Center (VA NSOC) utilizes the SolarWinds Network Performance Monitor (NPM) to monitor many areas of the VA Enterprise Network. This document focuses on two areas: 1) How SolarWinds monitors the Citrix NetScaler equipment deployed on the Enterprise Network, and 2) a description of the creation and function of the NetScaler Application Pool Monitors as they are created on the SolarWinds Platform in use at the VA.

1.1 Purpose

The purpose of this document is to provide as much information as possible to VA NSOC personnel to understand what is monitored within SolarWinds, to understand the additional development made to SolarWinds to accomplish this monitoring capability to the Application Pool Member level, and to duplicate this effort or make changes, if needed.

1.2 Scope

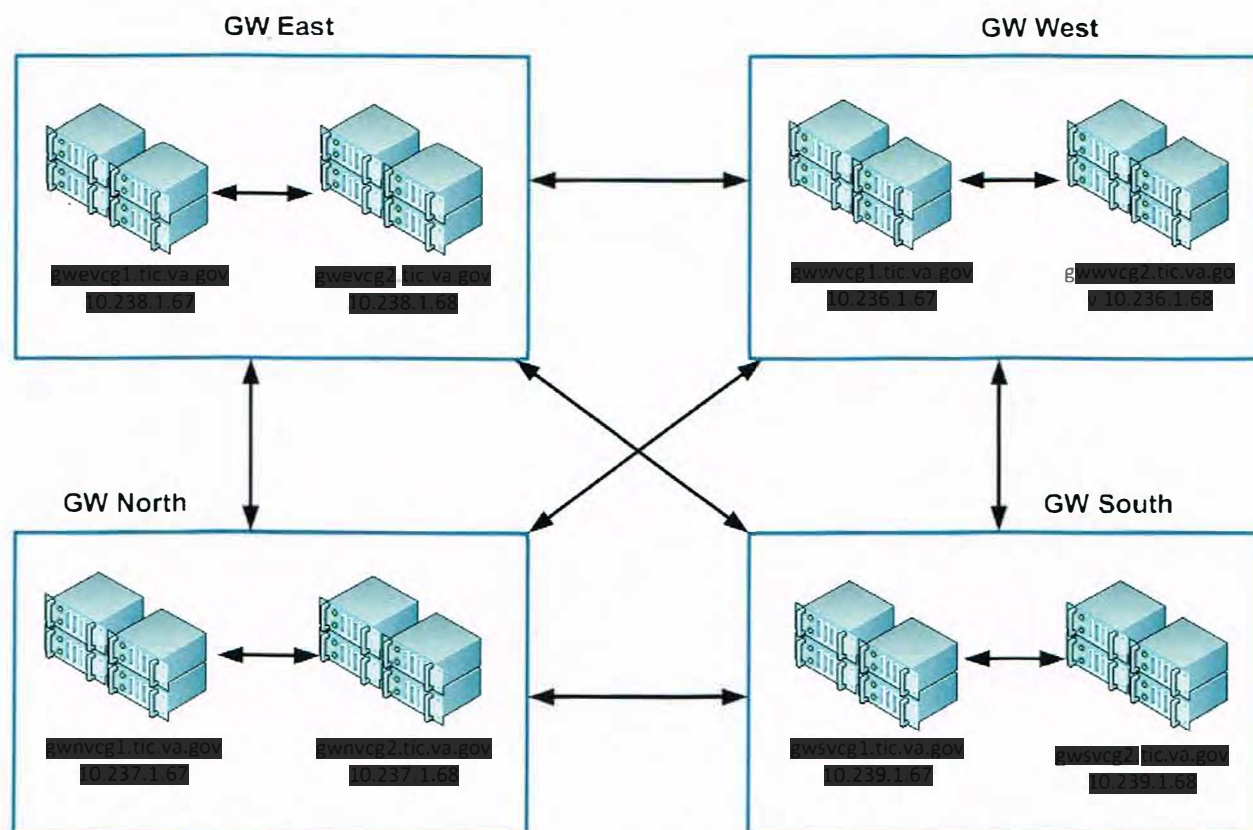
The scope of this document focuses on two groups – the VA NSOC Visibility Team and the Enterprise Network and Security Transition Team (ENSTT). The primary focus is on the Visibility Team members for cross-training purposes. The scope also pertains to ENSTT in their understanding how this information is created within SolarWinds, what is occurring behind the scenes, and what the statuses mean when being viewed.

2. NetScaler Overview within the NSOC

The VA NSOC was tasked by the IT Chief Information Officer (CIO) to provide Visibility to Everything (V2E). To accomplish this tasking and goal, the VA NSOC Visibility Team decided to add the monitoring of the NetScaler Application Pools to SolarWinds. These sections and the following graphics provide some detailed information on how the NetScaler environment is set up within the VA Enterprise Network and the configuration of data to accomplish this monitoring.

Sensitive But Not Classified

Gateway Netscaler Replication Overview



There are 2 Netscaler devices in each of the 4 GWs = 8 physical devices. All 8 Nodes are ACTIVE – ACTIVE and know all Application Servers on every single node. These devices are monitoring 44 Application Pools.

Sensitive But Not Classified

Figure 2-1: NetScaler Replication across the VA Enterprise Network

2.1 NetScaler Overview

There are two NetScaler devices in each of the four Gateways (GWs), making eight physical devices. Each NetScaler device (will use Node moving forward) has an active-active connection between each of the Nodes at each GW and between all four GWs. All eight physical Nodes

have all the same information; the only difference is where the location of a specific Application Pool Member may be located.

For instance, in using a filing cabinet analogy, Node 1 at GW East may have a specific member located in drawer two, file folder number eight from the front, but that same member may be located in Node 2 in GW West in drawer five, file folder 2 from the front.

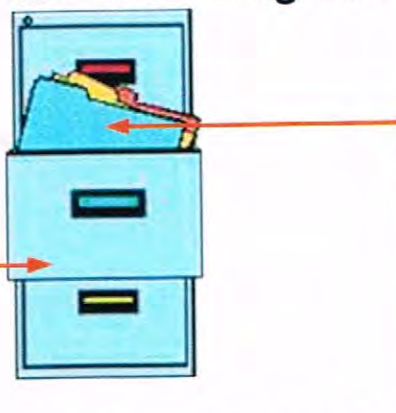
Again, the information is the same, and the members are all the same; only the location of that information is different between GWs.

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NetScaler Overview – File Cabinet Analogy

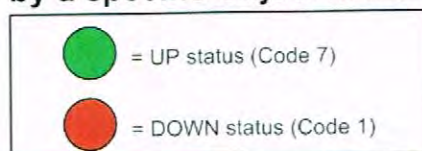
NetScaler Node = the Filing Cabinet

**App Pool = the Filing
Cabinet drawer**



**Node Element (IP Address) = the
file folders in a specific drawer**

The status of that node/IP address is contained within the file folder and specified by a specific Object Identifier (OID)



Sensitive But Not Classified

Figure 2-2: NetScaler Overview – File Cabinet Analogy

2.2 SolarWinds Retrieval Process – Object Identifier Status

SolarWinds NPM is configurable to monitor many different network components from the “out-of-the-box” configuration. The monitoring is accomplished down to the Object Identifier (OID) level, took a high-level of understanding of the two systems, what monitoring capabilities were needed, several weeks of code manipulation, and processing.

NOTE: This specific status coding within SolarWinds is peculiar to the VA NSOC and was deemed infeasible by SolarWinds engineers.

The high-level overview of how this retrieval process works is displayed in the graphic below.

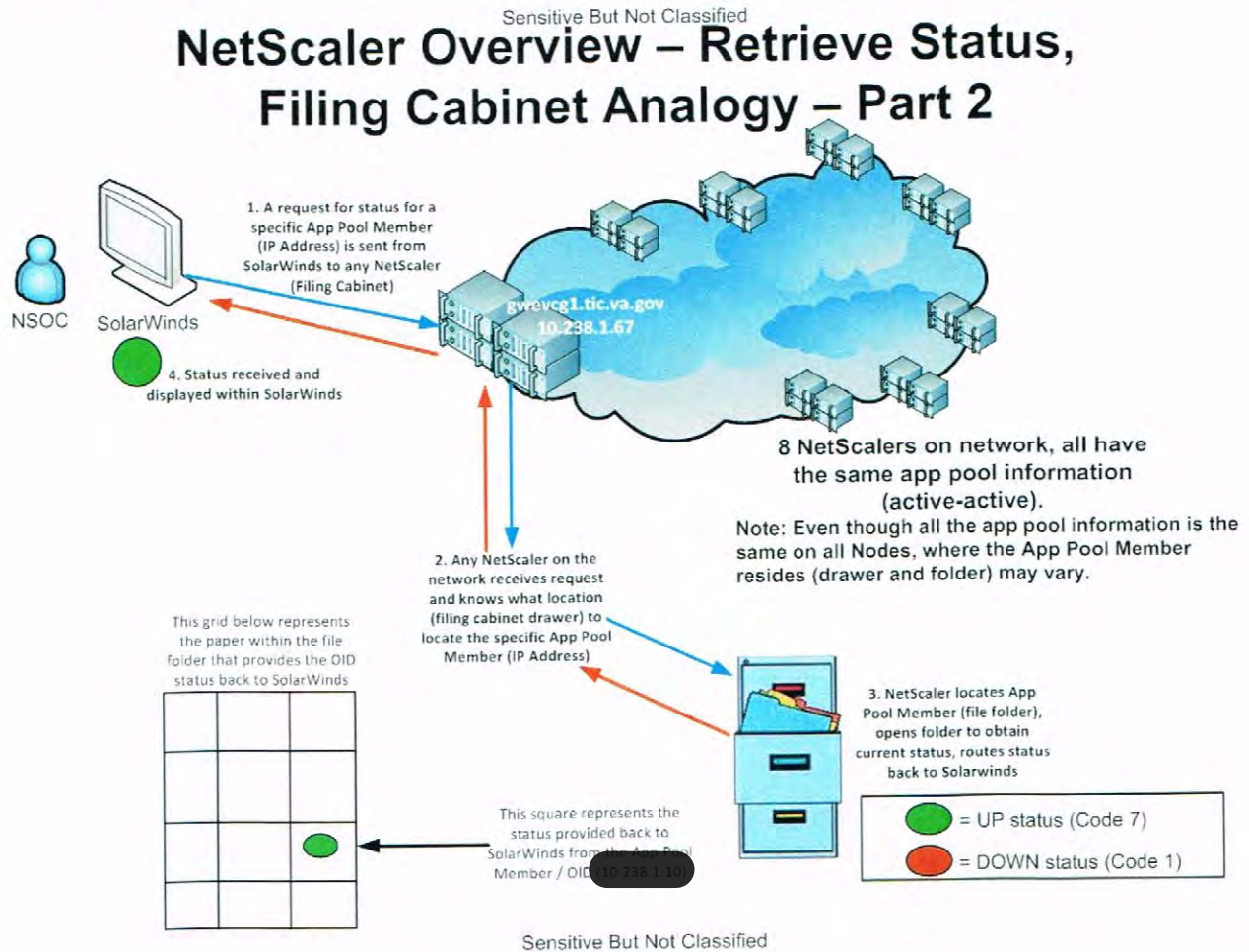


Figure 2-3: SolarWinds Retrieval Process – OID Status

1. A request for status for a specific App Pool Member (IP Address) is sent from SolarWinds to any NetScaler (Filing Cabinet).
2. Any NetScaler on the network receives request and knows what location (filing cabinet drawer) to locate the specific App Pool Member (IP Address).
3. NetScaler locates App Pool Member (file folder), opens the folder to obtain current status, and routes status back to SolarWinds.
4. Status of the App Pool Member is received and displayed by SolarWinds.

3. NetScaler Monitor and Member Nodes

The following two sections provide the steps for creating a new NetScaler monitor, and how to add member nodes to the NetScaler Farm template. The steps for creating a new NetScaler monitor are a two-part process. If you need to add nodes to an existing monitor, skip to Section 3.2, and complete the related steps.

3.1 Steps for Creating a New NetScaler Monitor Application

In creating a new NetScaler monitor or adding a member node to an existing NetScaler Farm, the “farm” name and Internet Protocol (IP) address(es) will be provided. An example of a “farm” name would be `XA_XML_FPO_VHAFPOCSF001_8080_svcg`. The IP addresses provided will be the members of the NetScaler Farm and will be used to name the Simple Network Management Protocol (SNMP) component monitors within Section 3.2.

NOTE: The steps below take for granted that you have administrative rights to the SolarWinds application and have a moderate-to-high understanding of this tool.

1. Navigate to the **Manage Applications** page within the SolarWinds tool.
2. Access the **Manage Application Monitor Templates** page, and click on the **Application Monitor Templates** tab.
3. Click on the **Create New Template** button near the top of this Application Monitor Templates tab.

Manage Application Monitor Templates

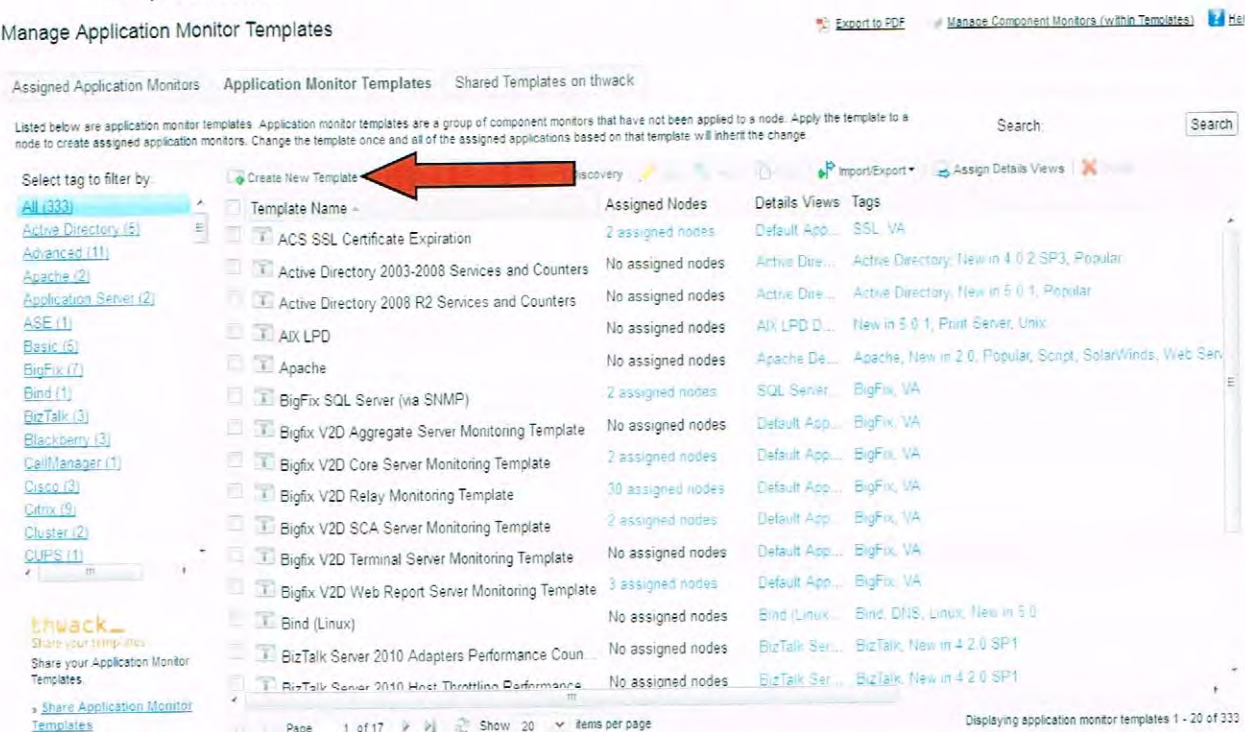


Figure 3-1: Manage Application Monitor Templates Window

4. Type in the **NetScaler Farm** name provided for monitoring into the **Template Name**. In this example, we will use `XA_XML_FPO_VHAFPOCSF001_8080_svcg`.

NOTE: Tags – It is acceptable to input additional values for this NetScaler Farm name, but you must use at least these two values – VA and NetScaler.

- Click on **Add Component Monitors** button, and input as many component monitors under the farm for which you have IP addresses.

Once you have input all the required Component Monitors (IP addresses), your final template would look similar to the one below. Remember – a SNMP Component Monitor will be displayed for each IP address that you have added.

Edit Application Template

Template Name: XA_XML_FPO_VHAFPOCSF001_8080_svcg

Template Applied to:

Description:

Tags: VA, NetScaler

Custom Application Details View: No, use Default Application Details View

Polling Frequency: 300 seconds

Polling Timeout: 300 seconds

Advanced

Quick Credentials (optional)
Are most or all of your component credentials the same? Use quick credentials to pre-fill all component credential fields. You can always change component credentials individually.

SNMP Monitor

Add Component Monitor

Submit Cancel

Test node not specified

Set test node

Test All

Test Rename Delete

Figure 3-2: Example of Final NetScaler Farm and SNMP Monitor Template

- Click the **Submit** button to save this new NetScaler Farm template.

3.2 Adding Member Nodes to the New NetScaler Farm Template

Note: The steps listed below apply if you are adding IP addresses to an existing NetScaler Farm Template. If the IP addresses were added from the Template area, SolarWinds will apply the new monitors when they are submitted. The IP addresses will not need to be applied again to the nodes.

- Navigate to the **Application Monitor Templates** tab within the SolarWinds tool.
- Locate the **NetScaler Farm name** that needs to be modified, or locate the farm name that was just created in Section 3.1 (XA_XML_FPO_VHAFPOCSF001_8080_svcg).
- Click on the box next to the farm name, and click **Edit** in the toolbar.

This will populate the *Edit Application Template* window as shown below.

(This is an example of an existing NetScaler Farm that is being modified. If you are editing the NetScaler Farm created in Section 3.1, the template will be empty.)

NetScaler Server Group XA_XML_FPO_VHAFPOCSF001_8080_svcg template

Template Name: [Redacted]

Template Applied to: [Redacted]

Description: [Redacted]

Tags: NetScaler, VA

Custom Application Details View: No, use Default Application Details View

Polling Frequency: 300 seconds

Polling Timeout: 300 seconds

+ Advanced

+ Quick Credentials (optional)
Are most or all of your component credentials the same? Use quick credentials to pre-fill all component credential fields. You can always change component credentials individually.

SUBMIT CANCEL

+ [Redacted] Test node not specified Set test node Test All

+ [Redacted] Test node not specified Set test node Test Rename Delete

Add Component Monitor

SUBMIT CANCEL

Set test node Test All

Test node not specified Set test node Test Rename Delete

Test node not specified Set test node Test Rename Delete

Add Component Monitor

SUBMIT CANCEL

Set test node Test All

Figure 3-3: Example of an Existing NetScaler Farm – Edit Mode

- Click the + (plus sign) next to the *SNMP Component Monitor* to add in a new component monitor or to modify an existing monitor. The corresponding component monitor edit window will display.

SUBMIT CANCEL

+ [Redacted] Test node not specified Set test node Test All

+ [Redacted] Test node not specified Set test node Test Rename Delete

Add Component Monitor

SUBMIT CANCEL

Set test node Test All

Test node not specified Set test node Test Rename Delete

Test node not specified Set test node Test Rename Delete

Add Component Monitor

SUBMIT CANCEL

Set test node Test All

Figure 3-4: SNMP Component Monitor Window

Figure 3-5: SNMP Component Monitor Edit Window

Rename the monitor using one of the IP Addresses belonging to that farm

5. Click on the **Rename** button at the top right side of this window.

6. Input the desired IP address into the text field of the *Rename* window that appears and input the same IP address into the *Description* field.

7. *Static Warning Threshold* – Click on the drop-down arrow next to the *Static Warning Threshold*, and change to **less than or equal to**. Input **6.00** within the value field next to the drop-down selection.
8. *Statistic Critical Threshold* – Click on the drop-down arrow next to the *Statistic Warning Threshold*, and change to **less than or equal to**. Input **1.00** within the value field next to the drop-down selection.

Enable Component: Yes

Object Identifier (OID):

Custom SNMP port (set to 0 to use default SNMP port): 0

Convert Value: Yes, convert returned value. The conversion was introduced by Nessoside & approved. [a Learn more](#)

Statistic Warning Threshold: less than or equal to 6.00

Statistic Critical Threshold: less than or equal to 1.00

User Notes

How to obtain the Object Identifier (OID) string

The final value required is the OID string and is a little more difficult to obtain. The following table provides the GW Name and the related primary and secondary NetScaler devices.

Table 3-1: Gateway and SolarWinds Server Name

Gateway Name	SolarWinds Server Name
GW – North	• [Redacted] • [Redacted]
GW – South	• [Redacted] • [Redacted]
GW – East	• [Redacted] • [Redacted]
GW – West	• [Redacted] • [Redacted]

9. Locate the *SolarWinds Engineer Toolkit* that is loaded on one of the above listed SolarWinds servers.
10. Open up the **Management Information Base (MIB) Browser tool**, and input the IP address of one of these NetScaler devices (SolarWinds Server Name). For this document and steps, the IP address [Redacted] is being used, which is [Redacted]. You may need to check additional NetScalers using this tool – there are a total of eight NetScaler servers.
11. Be sure you are able to query the IP address via SNMP with the community string entered. The root OID that will be used for the serviceGroupMember Table is [Redacted].
12. Within the MIB browser tool, click on **MIB from the top toolbar**, and then click the **Search MIB Database** button.
13. Type the value [Redacted] and then click on the **Show in MIB Tree** button.
14. Click the **Get Table** button.  This action will take approximately five minutes to completely the query and load this table. The entire window with the downloaded table will appear.

Figure 3-7: Export – Export to Excel Spreadsheet Option Window

16. Once the spreadsheet has been exported to a chosen file location, open the Excel spreadsheet.
17. Locate the *svcGrpMemberPrimaryIPAddress* column. Conduct a **Find** within this same column for the IP address that needs to be configured.

There will be several entries. One entry will have an associated *svcGrpMemberPrimaryPort* of 137 – ignore that entry line. The other entry line will have the correct port number. This port number should be provided in the Farm name – [REDACTED]

NOTE: The IP address and the associated correct port number within this naming convention.

18. Highlight this entire row, and use the bottom scroll bar to look up the correct initial column INDEX for this row. (In this example, Row 179 is the target row). The displayed table is shifted one column to the right for display purposes only. The first column which contains the required OID string is not displayed below.

**Figure 3-8: Excel Spreadsheet Example OID String**

19. [REDACTED]

20. Prepend the above string with the value [REDACTED] The value string will now look similar to this:



NOTE: The value above represents the “*svcGrpMemberState*” entry of the “*serviceGroupMemberTable*” that was identified in a previous step.

21. Copy this entire string value, and then navigate back to the *NetScaler Farm template*.
22. Paste this entire string value into the OID field. Your template will now look similar to the window below, displaying the ending digits of this string value in the OID field.

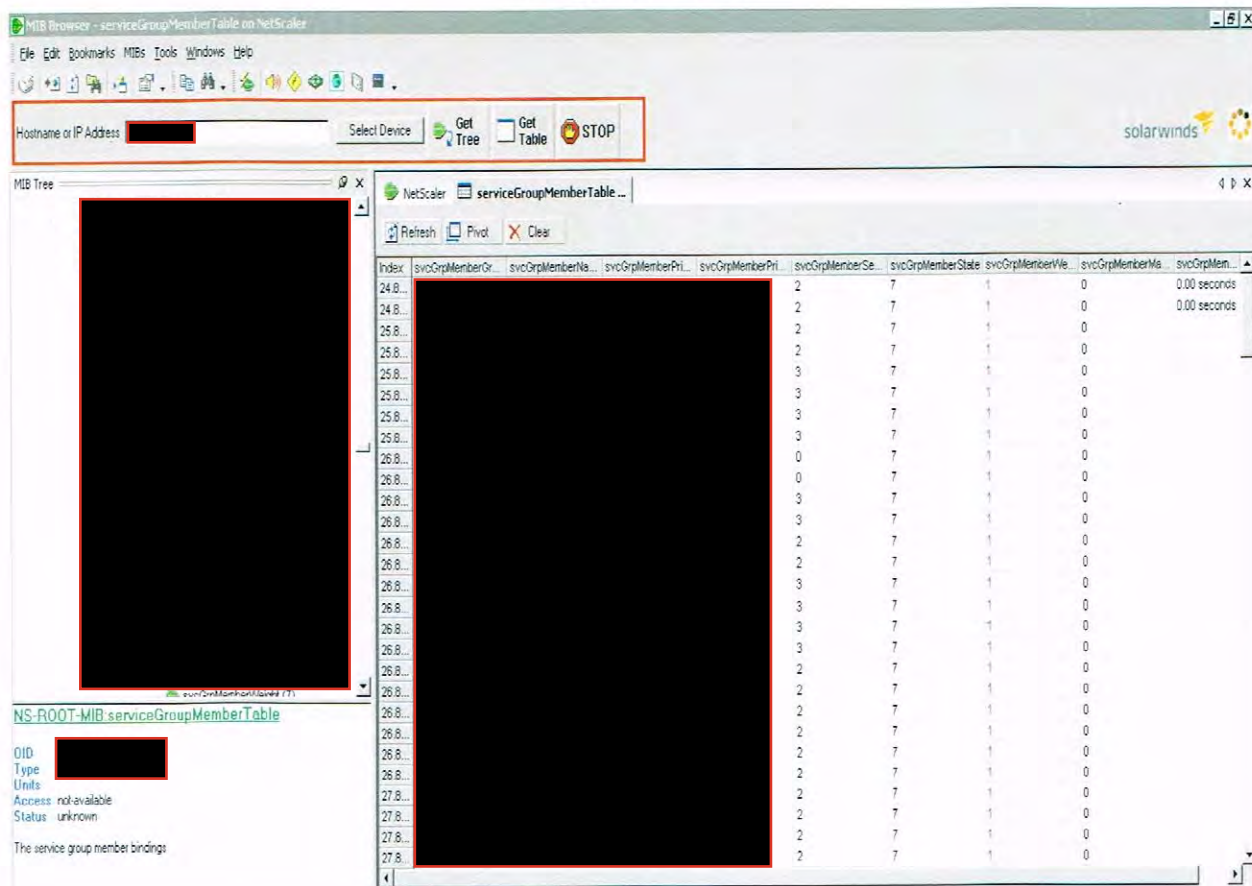
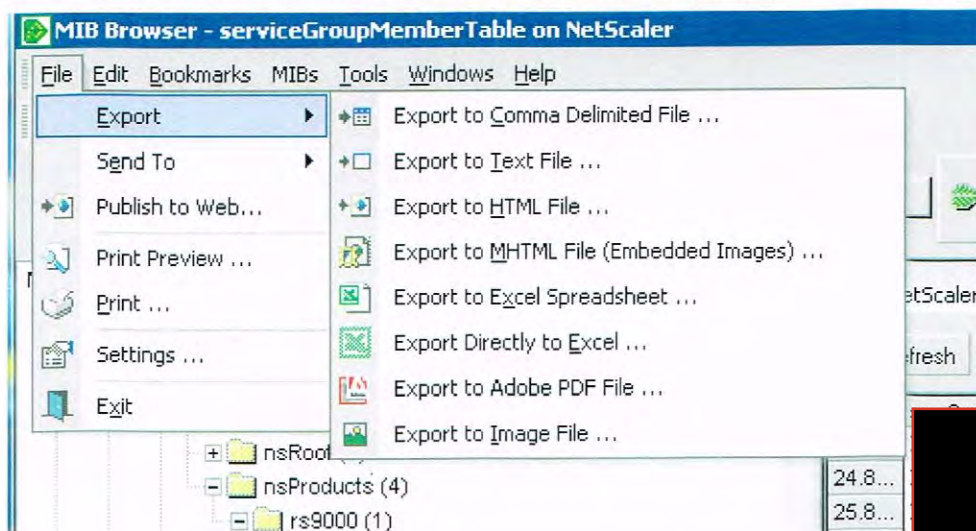


Figure 3-6: MIB Browser Window

NOTE: If you have multiple new NetScaler Farms and/or IP addresses to add to these farms, it may be more productive to export this table to an Excel spreadsheet, make the changes, and then import the spreadsheet back into SolarWinds.

15. In order to export this table to Excel, click on **File** in the upper-left corner, click on **Export**, and then select the **Export to Excel Spreadsheet...** option as shown below.



10.191.12.202 Test node not specified [Set test node](#) [Test](#) [Rename](#) [Delete](#)

Description 10.191.12.202

Component Type: SNMP Monitor

[About SNMP Monitor](#)
This component monitor uses the SNMP protocol to retrieve an entry in a Management Information Base (MIB) by its Object Identifier (OID), and returns the value of the entry as a statistic.
[more information](#)

Enable Component: Yes

Object Identifier (OID):

Custom SNMP port (set to 0 to use default SNMP port): 0

Convert Value: ☐ Yes, convert returned value.
The converted value will be used by thresholds & displayed [Learn more](#)

 **Statistic Warning Threshold:** less than or equal to

 **Statistic Critical Threshold:** less than or equal to

User Notes

Figure 3-9: SNMP Monitor Edit OID Window

23. Complete all of these above steps for each new *SNMP MONITOR* that needs to be created for the NetScaler Application Farm, and then submit your template.
24. Once your NetScaler Application Farm template is completed, you will need to assign the template to all of the GW SolarWinds Servers listed above in Table 3-1: Gateway and SolarWinds Server Name.

IMPORTANT:

If the component monitor fails (Red status within SolarWinds) on any of the NetScaler nodes – repeat the OID look-up process for that particular node(s) that was just added.

Add in another SNMP monitor with an identical IP address to the one that is failing, but use the complete OID string from the new lookup. Initially, add the identical IP address as “disabled.” Disable the monitor that is failing (Red status) on the affected nodes, and manually enable the new SNMP Monitor.

Appendix A – Acronyms and Definitions

Acronym	Definition
CIO	Chief Information Officer
ENSTT	Enterprise Network and Security Transition Team
GW	Gateway
IP	Internet Protocol
MIB	Management Information Base
NPM	Network Performance Monitor
OID	Object Identifier
SNMP	Simple Network Management Protocol
VA NSOC	Veterans Affairs Network Security Operations Center
V2E	View to Everything

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